BANOTH ABHINAV | 26IM61R06

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Percentage
2026–28	M.Tech, Operations Research and Data Analytics	IIT Kharagpur	Pursuing
2025	B.Tech, Mechanical Engineering	National Institute of Technology Calicut	6.53 / 10
2021	Class XII, TSBIE (MPC)	Sri Chaitanya Junior College, Hyderabad	92.8%
2019	Class X, CBSE	St. Peter's Central Public School	74.0%

PROJECTS

Robust Airport Gate Assignment under Delay Uncertainty | Oct'26 – Present *Python, Gurobi, scikit-learn*

- Built an **estimate-then-optimise** pipeline: gradient-boosting **quantile models** of arrival/departure delay (trained on 23,900 flights) feed a **stochastic MILP** (Şeker & Noyan, 2012) solved in **Gurobi**
- Generated 1,000 delay scenarios by residual bootstrap and reduced them to 30 with **k-means**; a scenario-aggregation reformulation keeps model size independent of the number of scenarios
- Added a **rolling-horizon** fix-and-optimise heuristic that warm-starts the exact MILP, cutting solve time ~4× (21.6 s to 5.2 s) at near-equal quality
- Over 500 out-of-sample simulated days, cut expected gate conflicts by **49%** and overlap minutes by **36%** vs. the schedule-based plan; conflict-free days rose from 33% to 56%

Gate Assignment IP – Chong & Nordin (2015) Replication | Aug'26 – Sep'26 *IBM CPLEX (OPL, DOcplex)*

- Formulated the integer program in **OPL and Python DOcplex**; reproduced all **3 published optimal solutions** exactly after identifying and correcting **2 data errors** in the paper
- Proved **total unimodularity** of the constraint matrix and mapped **8 distinct optimal assignments** across the full range of objective weight ratios

Bearing Predictive Maintenance with an LLM Assistant | Oct'26 *Python, scikit-learn, LLM*

- Processed the **NASA IMS run-to-failure** vibration data (3,140 recordings, 20 kHz) into 10 time-domain and **envelope-spectrum** features (Hilbert transform, bearing defect frequencies)
- An **Isolation Forest** trained only on healthy data flagged all 3 bearing failures **55–173 h in advance** with no false alarms under leave-one-bearing-out validation
- Packaged model outputs into a JSON evidence report that an **LLM** turns into plain-language maintenance advice for technicians

Customer Subscription Prediction – Replication Study | Oct'26 *R*

- Replicated 5 classifiers (DT, RF, LR, SVM, GBM) from a published study and exposed **target leakage**: without the leaked features, accuracy fell from 0.84 to the 0.73 majority baseline and ROC-AUC from 0.90 to ~0.50

SKILLS AND EXPERTISE

- **Optimization:** Gurobi (gurobipy), IBM ILOG CPLEX (OPL, DOcplex), AMPL; LP/MILP, duality, stochastic programming
- **Programming & Data:** Python (NumPy, Pandas, scikit-learn, Matplotlib, Seaborn), R, Django, Git/GitHub
- **Machine Learning:** Gradient boosting, quantile regression, random forest, isolation forest, SVM, k-means, logistic regression

COURSEWORK INFORMATION

- Operations Research
- Statistical Decision Modelling
- Recommender Systems in E-Business
- Safety Economics
- Applied Multivariate Statistics

POSITIONS OF RESPONSIBILITY

Student Representative | Industrial and Systems Engineering Society, IIT Kharagpur *2026 – Present*

AWARDS AND ACHIEVEMENTS

- Represented **NIT Calicut** at the **Inter-NIT Cricket Tournament three times**
- Runner-up (2nd place), **Kerala Cricket Premier League** (North Zone); played at Calicut district and North Zone Kerala levels

EXTRA CURRICULAR ACTIVITIES

- Badminton at a competitive level; trekking, open-water swimming, adventure sports, chess; hands-on natural farming
- Designed and deployed a personal portfolio website (abhistrikes.github.io) on GitHub Pages